

ORF307: Optimization

Precept: #7

Agenda

- Review of the Simplex method
- Optimality condition question
- Degeneracy and cycling
- Example

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Standard form of polyhedra

Feasible directions

Basic Feasible Solution (BFS)

Basic feasible direction d

Cost of improvement

Optimality conditions

Example: Simplex iteration (also known as Simplex pivot)

Let's see in detail the operations performed in a single iteration of the Simplex method.

We consider the minimization problem:

$$\min c^T x = -3x_1 - x_2$$

subject to:

$$x_1 + x_2 \leq 4$$

$$2x_1 + x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

Practice exercise

Compute the optimal solution of the LP by performing 3 Simplex iterations (pivots).

$$\min c^T x = -3x_1 - x_2$$

subject to:

$$x_1 + x_2 \leq 100$$

$$x_2 \leq 40$$

$$x_1 - x_2 \leq 30$$

$$x_2 - x_1 \leq 30$$

$$x_1, x_2 \geq 0$$

Figure 1 shows the feasible region of the LP.

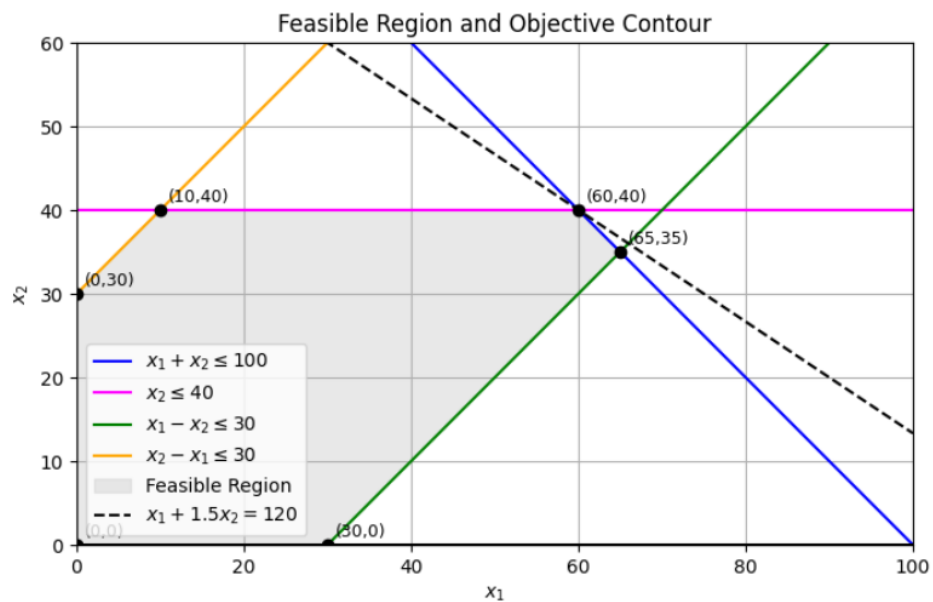


Figure 1: Visualization of the feasible region and objective contour of the practice exercise.